

Programme	B. Sc. STATISTICS				
Course Code	STA 7 CJ 402 (P)				
Course Title	PROBABILITY THEORY				
Type of Course	Major				
Semester	VII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practicum per week	Total Hours
	4	3	-	3	75
Pre-requisites	Basic Probability theory, Concept of convergence				
Course Summary	Understanding expectation and various celebrated theorems in classical probability theory.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Recall sequence of sets and its convergence.	R	F	Instructor-created exams
CO2	Explain the concept of field, sigma field, Borel sigma field and monotone class.	U	C	Practical Assignment
CO3	Describe various types of measures and explain its properties.	R	C	Seminar Presentation
CO4	Explain decomposition of distribution function, characteristic function and its properties.	Ap	P	Instructor-created exams / Home Assignments
CO5	Explain the monotone convergence Theorem, Fatou's Theorem and Lebesgue dominated convergence Theorem.	Ap	P	Writing assignments
CO6	Explain the concept of convergence in probability, Convergence almost surely, Convergence in distribution, Convergence in r^{th} mean and their inter-relations.	U	F	Group Tutorial Work
CO7	Illustrate the convergence of distribution functions, Helly-Bray Lemma and Helly – Bray theorem, and Levy continuity theorem.	An	P	Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I			10	20
	1	Definition of minimal sigma field, generated sigma field and induced sigma field		
	2	Random variables, Sigma fields induced by random variables, Vector random variables, limits of sequence of random variables.		
	3	Concept of measure space, finite measure, sigma finite measure, complete measure, counting measure and signed measure (Definition and examples only).		
	4	Probability space, General Probability space.		
	5	Induced probability space.		
II			12	15
	6	Decomposition of distribution functions, Distribution function of vector random variables, Correspondence theorem.		
	7	Expectation and moments, Properties of expectations.		
	8	Moments and inequalities		
	9	Characteristic functions, Properties, Inversion theorem		
	10	Characteristic functions and moments, Bochner's theorem (No proof required)		
	11	Independence of classes of events; Independence of random variables		
	12	Kolmogorov 0-1 law; Borel 0-1 law		
III			12	20
	13	Monotone convergence Theorem.		
	14	Fatou's Theorem		
	15	Lebesgue dominated convergence Theorem		
	16	Lebesgue-Stieltjes integral and its reduction to Riemann-Stieltjes integral and Riemann integral.		
	17	Statement and applications of Lebesgue decomposition and Radon-Nykodym theorem.		
IV			11	15
	18	Convergence in probability, Convergence almost surely		
	19	Convergence in distribution, Convergence in rth mean – their inter-relations- examples and counter examples.		
	20	Weak convergence		
	21	Helly-Bray Lemma and Helly – Bray theorem		
	22	Levy continuity theorem.		
V	Open Ended		30	
		Sequences of sets, limit supremum, limit infimum and limit of sets. Monotone sequence of sets. Fields, Sigma fields, Borel sigma field and monotone class. Hands-on-activities using Python/R. Open book problem solving exercises		

Text Books

1. B.R Bhat (1999). Modern Probability Theory, Wiley Eastern
2. Laha & Rohatgi (1979). Probability theory, Wiley New York
3. De Barra, G. (2000). Measure Theory and Integration, New Age International (P) Ltd, New Delhi.

References

1. **Ash R. B (2000)**. Probability and Measure Theory, Second edition. Academic Press.
2. **Billingsley P (1985)**. Probability and Measure, Second edition, John Wiley and Sons, New York.